

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

Nov/Dec 2013

EE-217 ENGINEERING ELECTROMAGNETICS

Date: 28.11.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define following:

[05]

- Coulomb's Law
- Divergence Theorem
- Gauss Law
- Electric field intensity
- Potential Gradient

(b) Derive the expression of E due to point charge

[02]

Q - 2 (a) If $\vec{A} = 10\vec{a}_x + 5\vec{a}_y + 5\vec{a}_z$ $\vec{B} = 5\vec{a}_x + 10\vec{a}_y + 10\vec{a}_z$

[05]

- I. a unit vector in the direction of $\vec{A} - \vec{B}$
- II. the magnitude of $\vec{A} + \vec{B}$.
- III. $|\left[(\vec{A} \times 3\vec{B}) / |(\vec{A} - 3\vec{B})| \right]| \cdot \vec{B}$.
- IV. Find Area of parallelogram made by the above two vectors.
- V. Vector component of A that is parallel to B

(b) Calculate resultant E at M (3,-4,2) in free space caused by charges

[02]

Charge $Q_1 = 2\mu\text{C}$ at point P1(0,0,0) and charge $Q_2 = 3\mu\text{C}$ at P2(-1,2,3)

(c) A point charge of 6nC is located at the origin in free space Find Vp if the point P is located at P (0.2, -0.4, 0.4) and at a.) V=0 at infinity b) V=0 at (1,0,0)

[07]

OR

Q - 2 (a) An infinitely long, uniform charge is located at y=3, z=5. If $\rho_L = 30\text{ nC/m}$

[06]

Find E at :

- 1.) origin
- 2.) at point P(0,6,1)
- 3.) at point Q (5,6,1)

- (b) Transform the vector $\vec{A} = 5 \vec{a}_x$ to spherical coordinates at the specified point [04]
 $B(x=-2, y=-3, z=-1)$
- (c) Derive the magnitude of potential difference V_{ab} between two points **a** and **b** located [04]
 in the field of line charge distribution. ρ_L

- Q - 3 (a) Derive the expression of Electric Field Intensity (E) at any point due to uniform line [05]
 charge distribution.
- (b) Obtain the expression of divergence of flux density D, for differential volume [05]
 element.
- (c) Explain what do you mean by potential and potential difference. [04]

OR

- Q - 3 (a) Derive the expression of energy or work done in moving a point charge in an external [05]
 electric field.
- (b) State and explain the divergence theorem. [04]
- (c) Derive the expression for finding Electric field intensity (E) from potential (V) giving [05]
 the gradient relation Between E and V.

SECTION - II

- Q - 4 (a) Define following: [05]
- 1.) BIO SAVART's LAW
 - 2.) Ampere's Circuital law
 - 3.) Curl Theorem-Statement
 - 4.) Faraday's Law of Electromagnetic Induction
 - 5.) Lorentz Force Law
- (b) State and prove Stokes theorem. [02]

- Q - 5 (a) Obtain the Expression of Magnetic field intensity $\vec{H}$ at any point due to conductor [07]
 carrying current I placed on Z axis from $-\infty$ to ∞ .
- (b) Differentiate between the Electric and Magnetic circuits. (Any 7 points) [07]

OR

Q - 5 (a) Derive the Expression of Curl of $\vec{H}$ or discuss the point form of Ampere's circuital Law. Also give the expressions for the same in various co-ordinate systems. [07]

(b) Discuss the application of Amperes circuital law for finding magnetic field intensity $\vec{H}$ due to at various radii of Coaxial Conductor. Discuss all the cases. [07]

Q - 6 (a) Define current & current density, and derive relationship between current density J volume charge density (ρ_v) and its velocity V . [06]

(b) Obtain the boundary conditions for dielectric materials. [04]

(c) Make a comparison between Conductors, Semiconductors & Insulators. [04]

OR

Q - 6 (a) Explain properties of perfect conductor. [06]

(b) Obtain the boundary conditions for magnetic materials. [04]

(c) Derive Maxwell's equation from Faraday's law in point form and integral form. [04]
